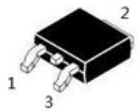
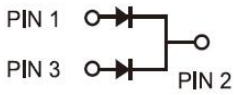
Leiditech

MBR20100CS

SCHOTTKY BARRIER RECTIFIER



TO-252/CS



FEATURES

- Low forward voltage
- High current capability
- High forward surge capability
- Low power losses，High efficiency
- Guarding for over voltage protection

APPLICATIONS

Low VF Schottky barrier rectifier are designed for high freqency, miniature switched mode power supplies such as adapters ,lighting and on-board DC/DC conerters

Primary Characteristic

I_O	2*10A
V_{RRM}	100V
I_{FSM}	250A
V_F	0.68V
$T_J$max	150℃
Assembly code	X X

MECHANICAL DATA

- **Case:** Molded plastic
- **Polarity:** As marked
- **Mounting Position:** Any
- **Molded Plastic:** UL Flammability Classification Rating 94V-0
- Lead free in compliance with EU RoHS 2011/65/EU directive
- Solder bath temperature 275℃ maximum,10s per JESD 22-B106

Maximum Ratings (Per Leg) at Ta=25°C unless otherwise specified

Characteristics		Symbol	Value	Unit
Maximum Repetitive Peak Reverse Voltage		V_{RRM}	100	V
Working Peak Reverse Voltage		V_{RWM}	100	V
Maximum DC Blocking Voltage		V_{DC}	100	V
Maximum Average Forward Rectified Current	Per Leg	I_O	10	A
	Total		20	
Peak Forward Surge Current,8.3 ms Single Half Sine-wave		I_{FSM}	250	A
Operating Temperature Range		T_J	150	°C
Storage Temperature Range		T_{STG}	-40 to +150	°C
Typical Thermal Resistance (Note1) TO-220AB,TO-263,TO-252 TO-220F		$R_{\theta JC}$	 2 4	°C/W

Note1: Thermal resistance from Junction to case per leg mounted on heatsink.

Electrical Characteristics (Per Leg) unless otherwise specified

Characteristics		Symbol	Value		Unit
Forward Voltage Drop(Note2)		V_F	Typ.	Max.	V
at I_F=3A	TA=25°C		0.69	-	
	TA=125°C		0.60	-	
at I_F=5A	TA=25°C		0.74	-	
	TA=125°C		0.61	-	
at I_F=10A	TA=25°C		0.82	0.85	
	TA=125°C		0.68	-	
Maximum Reverse Current at V_R=100V	TA=25°C	I_R	1.5	10	μA
	TA=125°C		1	-	mA

Note2:Pulse test: 300 μs pulse width, 1 % duty cycle

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RATINGS AND CHARACTERISTIC CURVES

FIG.1 MAXIMUM FORWARD CURRENT DERATING CURVE

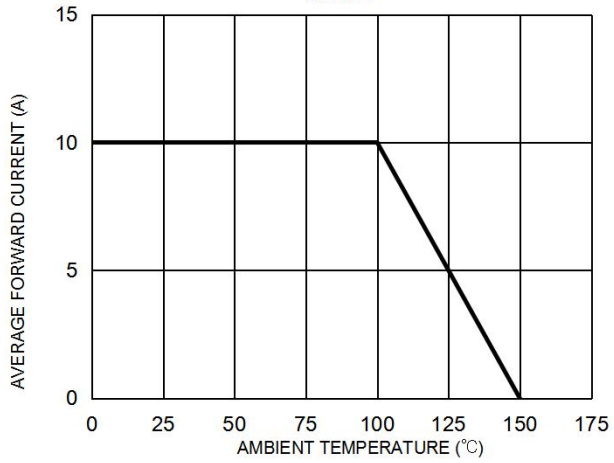


FIG. 2 MAXIMUM NON-REPETITIVE FORWARD SURGE CURRENT PER LEG

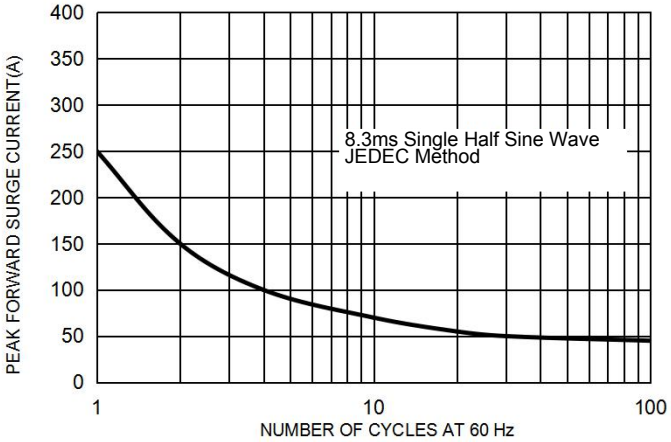


FIG. 3 TYPICAL FORWARD CHARACTERISTICS PER LEG

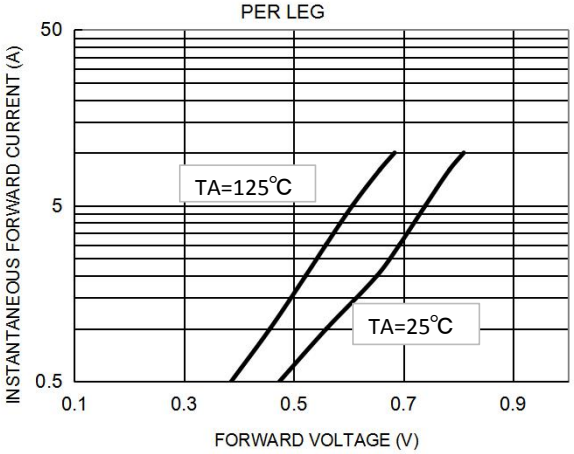
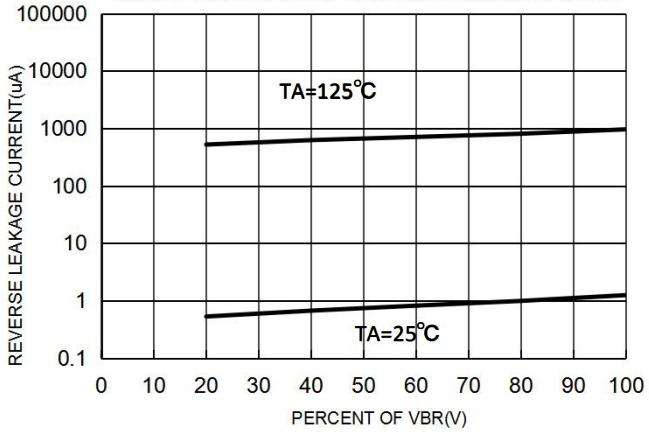


FIG. 4 TYPICAL REVERSE CHARACTERISTICS PER LEG



Ordering information

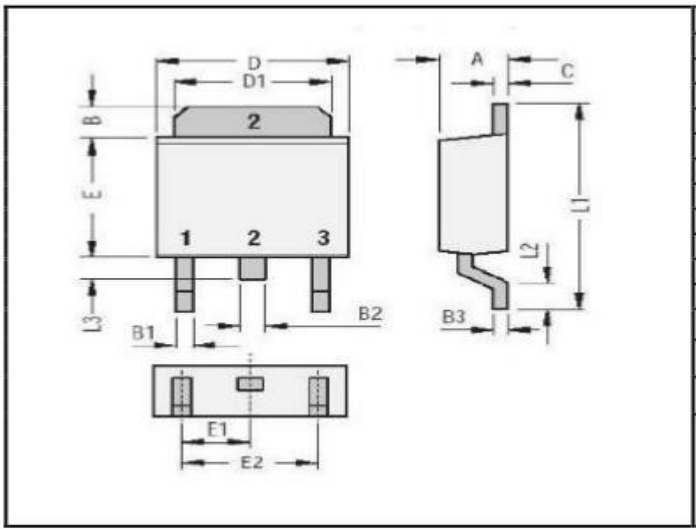
Part Number	Package	Unit Weight	Base Quantity	Delivery mode
MBR20100CS	TO-252	0.011oz(0.32g)	2500 pcs / reel	2500pcs/box 12500pcs/carton

Note: For Halogen Free molding compound, add H suffix to part number above.

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Package Outline Dimensions millimeters

TO-252



Dim.	Min.	Max.
A	2.1	2.5
B	1.3	1.7
C	0.4	0.6
D	6.4	6.7
D1	5.1	5.5
E	5.5	6.0
E1	Typ2.3	
E2	Typ4.6	
B1	0.6	0.8
B2	0.75	0.95
B3	0.35	0.45
L1	9.0	10.0
L2	1.4	1.6
L3	0.75	0.95

All Dimensions in millimeter

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